Formula 1

Telemetry Insights Visual Dashboard

MOTORSPORTS



Project Report By:

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Problem

Formula 1 is one of the most data-driven sports, yet driver impact is often overlooked

- Engineers want to identify optimal lap patterns
- Drivers benefit from objective performance feedback

What is hard about F1 telemetry analysis?

- Raw lap data is massive, multidimensional, and difficult to interpret
- Hard to identify patterns in driver performance manually
- Need automated way to group similar laps and identify driving styles

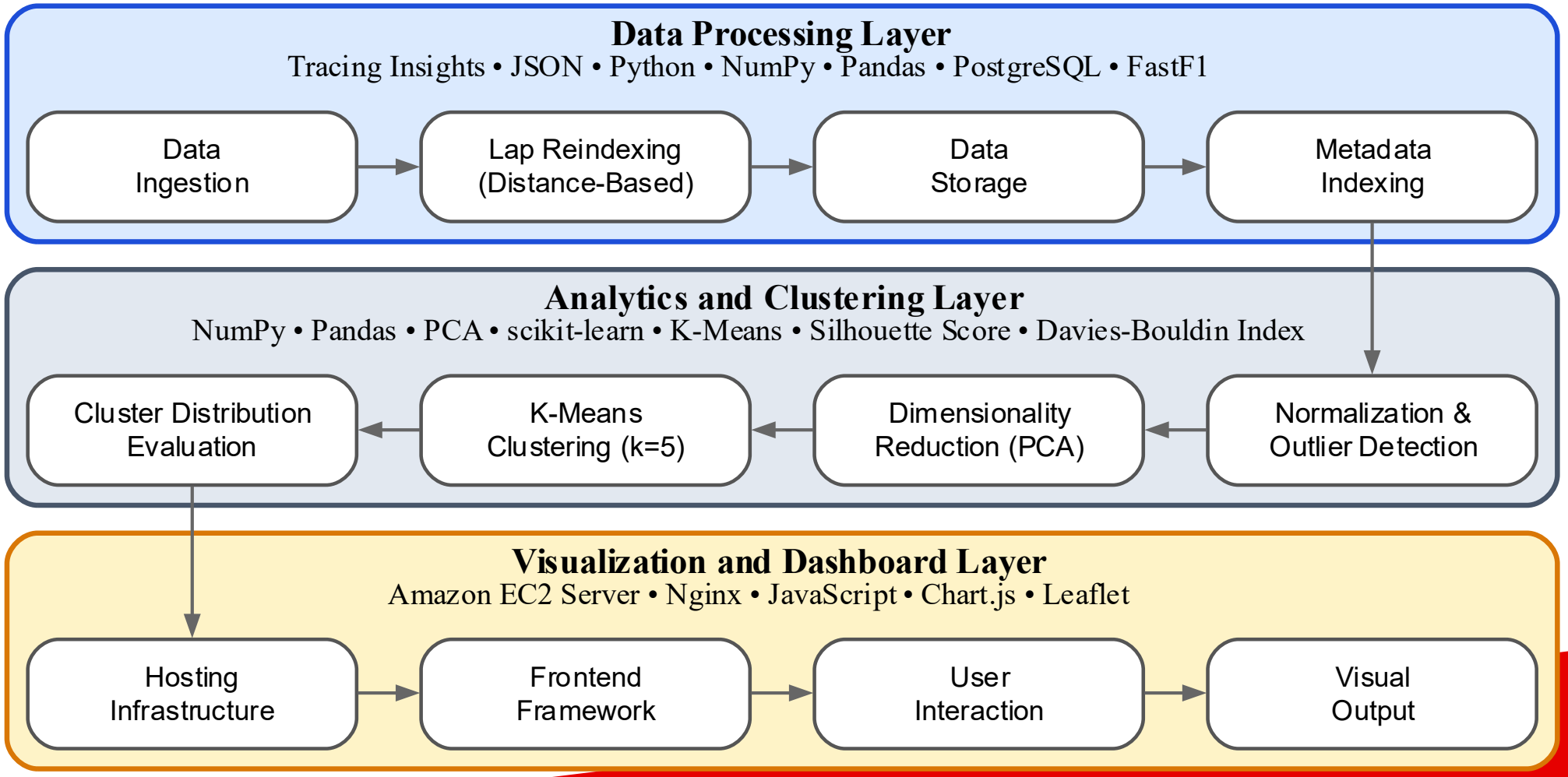


Objective

- Transform raw F1 telemetry into meaningful driver insights
- Identify performance patterns through clustering algorithms
    - Group drivers into "driving styles"
- Develop an interactive dashboard for intuitive data exploration
- Given the choice between two open source F1 data sources
    - Tracing Insights vs FastF1



Architecture



Initial Tracing Insights Data

Telemetry output for a single lap


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14 Total Telemetry Metrics

8 Selected Metrics

Throttle, RPM, Gear, Brake, Speed, Accel_x, Accel_y, Accel_z

Indexed by Relative Distance

Clustering Algorithm Selection

These options are the most common unsupervised machine learning algorithms

K-Means

SELECTED

✓ **STRENGTHS**

- Fast & scalable to 18M+ records
- Works well after PCA reduction
- Easy to tune cluster count (k)

✗ **LIMITATIONS**

- Requires pre-specifying k
- Assumes convex clusters

Silhouette: 0.0653 | Davies-Bouldin: 2.61

Hierarchical

TESTED

✓ **STRENGTHS**

- No need to pre-specify k
- Dendrogram reveals structure

✗ **LIMITATIONS**

- $O(n^2)$ memory — too slow for 18M+ records
- Cannot scale to full 2025 season
- Works best on smaller data sets

Ruled out: memory constraint

Gaussian

TESTED

✓ **STRENGTHS**

- Soft/probabilistic assignments
- Handles elliptical clusters

✗ **LIMITATIONS**

- Assumes Gaussian distributions
- Unstable on high-dim PCA space
- Higher compute cost

Ruled out: distribution assumption mismatch

DBSCAN

TESTED

✓ **STRENGTHS**

- No cluster count needed
- Auto-detects noise/outliers

✗ **LIMITATIONS**

- Struggles with varying density
- ε tuning sensitive — poor cluster consistency across tracks

Ruled out: inconsistent clusters per race

Four algorithms evaluated — K-Means selected for performance and interpretability

K-Means Clustering

Cluster Evaluation Methods:

- Elbow Method – identifies diminishing returns in cluster separation
- Silhouette Score – measures cluster cohesion vs separation
- Davies-Bouldin Index – evaluates cluster similarity (lower is better)

Our Findings after many experimental models

- K-Means at 6 clusters produced the most well-balanced clusters (~1-2 outlier groups per race)
- Clusters can vary wildly depending on the track
- Metrics can be misleading

Cluster Representation Examples

Cluster 1
High Aggression
High RPM, throttle, speed — peak-performance laps

Cluster 2
Controlled Push
Elevated throttle, measured braking — raceable pace

Cluster 3
Balanced
Avg. across all metrics — representative race lap

Cluster 4
Cautious
Lower throttle/speed — caution laps or degradation

Cluster 5
Conservation
Low RPM, heavy coasting — tire/fuel management

We leave it up to the user to explore the data and label the groups

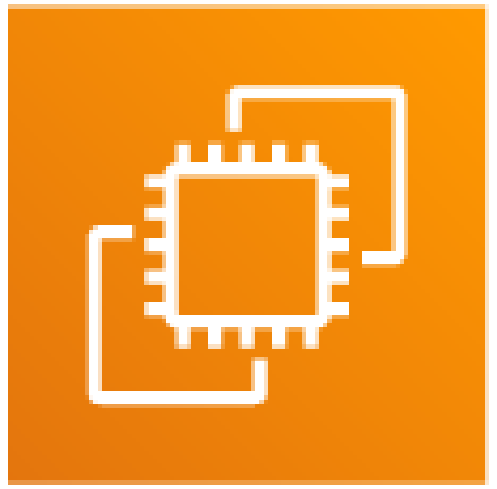
Deployment Stack

Accessed through a ***porkbun*** domain, served through ***NGINX***, hosted on an ***AWS EC2*** Instance.



porkbun

NGINX



Amazon EC2

Website



uark-team6-f1-visual-dashboard.site

Formula 1

Telemetry Insights Visual Dashboard

MOTORSPORTS



All the way back to the starting line...